Portfolio Optimization Model

Results & Application Summary

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What This Is

I built a Modern Portfolio Theory optimizer in Python that takes a list of stocks and solves for the portfolio weights that either maximize risk-adjusted return (Sharpe ratio) or minimize volatility. It uses 5 years of daily price data (2019-2024) across a 20-stock universe spanning tech, financials, energy, healthcare, consumer names, and two broad index funds (SPY, QQQ). I also added a realistic position cap (no more than 25% in any one stock) and a transaction cost model, since an optimizer with no limits will happily dump most of your money into 1-2 stocks, which isn't how anyone should actually invest.

Universe Tested

AAPL, MSFT, GOOGL, AMZN, NVDA, META, JPM, BAC, XOM, CVX, JNJ, PFE, KO, PG, WMT, DIS, V, HD, SPY, QQQ

Results

Max Sharpe Portfolio — Unconstrained

Ticker	Weight
AAPL	28.2%
KO	37.6%
WMT	10.9%
SPY	15.8%
QQQ	7.5%

Sharpe Ratio: 1.39

Max Sharpe Portfolio — Constrained (25% cap per position)

Ticker	Weight
AAPL	25.0%
KO	25.0%

SPY	20.6%
WMT	16.3%
QQQ	9.8%
PFE	3.3%

Sharpe Ratio: 1.37 (a 0.014 drop from unconstrained — the real cost of forcing diversification)

Minimum Variance Portfolio

Ticker	Weight
SPY	25.0%
BAC	28.7%
CVX	16.0%
WMT	9.3%
PG	8.2%
MSFT	6.5%
QQQ	5.7%
NVDA	0.6%

Return: 11.5% | Volatility: 16.3% (the lowest-risk combination the optimizer could find)

What $10,000 Actually Looks Like

Using the constrained (25% cap) portfolio, here's what a $10,000 allocation comes out to in real dollars:

Ticker	Dollar Amount
AAPL	$2,500.00
KO	$2,500.00
SPY	$2,057.69
WMT	$1,631.36
QQQ	$984.03
PFE	$326.92

Transaction Costs

Rebalancing $10,000 from an equal-weight starting point into the constrained target portfolio costs about $14.35 at a 0.1% cost per dollar traded. That knocks the Sharpe ratio down slightly, from 1.3697 to 1.3640. It's a small number here, but it's a real one, and it would add up fast if you rebalanced constantly instead of just occasionally.

What This Actually Means

The unconstrained version wanted to put almost 66% of the money into just two stocks (AAPL and KO), even though it had 20 to choose from. That's not a bug — it's just what unconstrained Sharpe maximization tends to do, since it's extremely sensitive to whatever expected returns you feed it, and it doesn't care about looking diversified, only about the math. That's a well-known real-world weakness of the basic Markowitz approach, and it's exactly why I added the 25% cap.

The min variance portfolio landed on SPY as its single biggest position, which makes sense once you think about it: SPY is already 500 stocks bundled together, so it's the least volatile individual thing in the whole list. The optimizer isn't being clever here, it's just correctly noticing that a pre-diversified asset is hard to beat on volatility alone.

The gap between the unconstrained Sharpe (1.39) and the constrained Sharpe (1.37) is basically the price you pay for not putting all your eggs in two baskets. That's a real, quantifiable tradeoff between "mathematically optimal" and "actually reasonable to hold."

How To Actually Use This

If I wanted to use this for real: pick a universe of stocks I'm comfortable holding, decide on a max position size I'm okay with (25% is what I used, but that's adjustable), run the constrained optimizer, and the output tells me exactly how much money to put into each name for whatever amount I'm investing. The transaction cost check tells me roughly what it'll cost to actually get into that position from wherever I'm starting, so I'm not surprised by fees eating into the return.

The honest way to think about this: it's a starting point for how to size positions given a set of stocks you've already decided you like, not a tool that tells you which stocks to pick in the first place. It also assumes the future looks like the past 5 years, which is never guaranteed — expected returns and correlations shift over time, so this is something you'd want to re-run periodically, not treat as a set-it-and-forget-it answer.

Limitations, Being Honest About It

• Expected returns are estimated from 5 years of trailing history, which is a noisy, backward-looking estimate of what a stock will actually return going forward.

• No shorting is allowed, so every weight is between 0% and the 25% cap. Real portfolios sometimes use short positions, which this doesn't model.

• Transaction costs are a flat 0.1% per dollar traded, a simplification of real costs like bid-ask spread and market impact, which actually vary by stock and trade size.

• This model treats risk purely as volatility (standard deviation). It doesn't account for tail risk, drawdown risk, or correlations breaking down during a crash, which is when diversification often matters most and works least well.